

Introduction to Regression and Linear Classification and Their Implementation in Python

Write a program in Python and perform the following tasks:

1- Consider the student grades file used in Assignment 1. Using the [Linear_model.LinearRegression\(\)](#) routine available in the [scikit-learn](#) library, fit a linear model to the above data. Use the values of the columns that specify the grades for the course components as the input (variable X), and the values in the total column as the output (yout). Finally, find and display the weight assigned to each of the specified course components.

Again, use the `read_excel()` routine in the [pandas](#) package to read the information in this file. Also study [DataFrame](#) in this package.

2- Put the values of the pass column in the input data into the variable t. Then, using the least-squares methods `RidgeClassifier()` (without regularization), stochastic gradient descent `SGDClassifier()` regression, and logistic regression `LogisticRegression()` available in `sklearn.Linear_model`, perform classification using the data (X, t). Then calculate and display their recall, accuracy, precision, and confusion matrix using [sklearn.metrics](#). Which of these three performs better?

3- Put the data values corresponding to the midterm-exam and final-exam columns into another variable named X2. Then classify the dataset (X2, t) using logistic regression and obtain the resulting decision boundary. Display the decision boundary on a scatter plot of the X2 data. In the scatter plot, show the points belonging to each class in a different color.

4- Using the least-squares method, obtain the decision boundary for the dataset (X_2 , t) using **code that you write yourself** and display it on a scatter plot of the X_2 data. Note that, to do this, X_2 and t must be expanded and 1-of-2 encoding must be used. Is the decision boundary obtained the same as the boundary from Exercise 3? Which one is better? In the scatter plot, show the points belonging to each class in a different color. (If necessary, you may use the numpy package and `numpy.linalg`.)

Submit the answers to the above questions and the results of your experiments in a PDF-format report together with your programs.

Prepare and submit a short video explaining the execution of your program.

Please pay attention to the following points:

- Assignments can be submitted up to two days after the deadline; 10% of the grade is deducted for each day of delay.
- Avoid colloquial writing, as it will result in a grade deduction. Do not include handwriting or images of handwritten material in your report.
- Only assignments submitted through the system will be graded. Under no circumstances should you email your assignment.
- No credit will be given for assignments that are similar to each other.

Good luck

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